

GenAI Training Day 5

Guardrails

LLM Output Quality

- LLM output directly impacts users and businesses
- Poor output creates legal, ethical, and reputational risk
- Production systems require predictable, controlled behavior

What Is LLM Output Quality?

- Output quality is multi-dimensional
- Accuracy alone is not enough
- Must balance correctness, safety, speed, and consistency

Key Quality Dimensions

- **Accuracy:** factual correctness & grounding
- **Latency:** response time for real-time vs batch use
- **Toxicity:** hate, violence, sexual or harmful content
- **Bias:** unfair assumptions or unequal treatment
- **Drift:** behavior changes over time or model versions

Why Guardrails Matter

- LLMs are probabilistic, not value-aware
- Safety is not inside the model
- Guardrails surround and filter model behavior
- Prevent harmful, unsafe, or undesired outputs

Types of Guardrails

- Content safety
- Data leakage prevention
- Bias control
- Hallucination control
- Output format enforcement

Content & Data Safety

- Block hate speech, violence, self-harm, extremism
- Prevent PII, secrets, and confidential data leakage
- Silence is safer than false confidence

Hallucination Guardrails

- Require citations or supporting context
- Reject unsupported answers
- Force “I don’t know” when necessary

Prompt Injection & Jailbreaks

- Users may try to override system instructions
- Attacks include role-play, encoding, indirect injection

- **Defenses:**
 - Input validation and sanitization
 - Jailbreak detection
 - Output re-checking

Retry & Recovery

- Automatic retries on failures or low confidence
- Adjust prompts, temperature, or timeouts
- Improves stability and reliability

LLM Evaluation

- Manual testing does not scale
- Models drift over time
- Continuous evaluation is mandatory
- Examples include **RAGAS**, **LangSmith**

Evaluation Approaches

- **LLM as Judge:** fast, scalable, regression testing
- **Human as Judge:** highest quality, slow, expensive
- **Human-in-the-Loop:** feedback and approval workflows

Evaluation Metrics

- Accuracy, relevance, completeness
- Faithfulness to context
- Safety, tone, and formatting